

WAP Manager & Dynamic Funnel Engine: End-to-End Client & Technical Manual

Important

This document serves as an exhaustive, step-by-step master reference for creating, configuring, customizing, executing, and auditing campaigns on the unified **WAP Manager (html-canvas)** platform. It details every architectural decision, backend mechanism, frontend editor binding, and logging system so you can confidently present and defend the platform to any client or team member.

1. Core Architecture & Philosophy: Why a Single Engine?

The Old Model (Fragmented Setup)

In legacy setups (such as [safwap-server-backup] (file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/safwap-server-backup)), every new campaign or telecom operator required:

- Deploying a separate application codebase or hosting static HTML files.
- Hardcoding operator callback URLs (**DigitalTraffic.sp_CPACallBack**) directly in backend code.
- Fragmented logging in flat **.log** files on disk.

The New Unified Model (**html-canvas**)

The modern platform uses a **Single Universal Multi-Tenant Runtime Engine**:

- **Zero-Code Deployment:** Unlimited campaigns run on one central engine ([SubscriptionPage.jsx]

(file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/frontend/src/pages/SubscriptionPage.jsx) + [flow.service.js] (file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/flow/flow.service.js)).

- **Dynamic Routing:** When a subscriber opens a link like `/subscription?country=India&operator=Jio&campid=101`, the system dynamically resolves the campaign, loads its custom GrapesJS HTML/CSS pages, fires configured partner APIs, and tracks analytics.

Public Consumer Link
(/subscription?
country=..&operator=..&campid=..)

Fastify Flow Gateway
(/flow/page)

Redis Cache (Flow Config /
API Config)

Database (campaigns,
campaign_pages,
api_configs)

Shadow DOM Runtime
Container
(SubscriptionPage.jsx)

User Clicks Action
(SUBSCRIBE / CONFIRM)

Flow Transition Gateway
(/flow/transition)

Partner APIs (Blocklist ->
Check Sub ->
Charge/Subscribe)



2. Step 1: Campaign Creation & Provisioning

2.1 Creating a Campaign

1. Navigate to **Campaigns Dashboard** (</campaigns>) in the Admin Panel.
2. Click **Create Campaign**.
3. Fill in the primary metadata:
 - **Name**: Campaign title (e.g. *Jio Games India*).
 - **Country**: Target country (e.g. *India, UAE, Saudi Arabia*).
 - **Operator**: Target telecom operator (e.g. *Jio, Etisalat, STC*).
 - **Service ID**: Billing identifier assigned by the carrier/partner.
 - **Verification Mode**: Choose from 4 engine modes ([HEADER_INJECTION](#), [OTP_ONLY](#), [BOTH](#), [NONE](#)).

Note

Reason for Operator/Country Binding: Telecom charging gateways require specific headers and endpoints tailored per operator/market. Binding [country](#) and [operator](#) guarantees clean URL routing and audit isolation.

3. Step 2: Understanding Verification Modes & Flow Graph

The engine's decision tree is powered by [flow-engine.service.js]
(file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/flow/flow-engine.service.js [#L3-L45](#)):

Verification Mode	Behavior & Entry Sequence	Primary Use Case
HEADER_INJECTION	Checks header for MSISDN (X-MSISDN). If found → proceeds directly to CONFIRM or THANKYOU . If missing → routes to ERROR page.	Direct 1-Click WAP / Mobile Data traffic where telco injects numbers.
OTP_ONLY	Skips auto header checks → requires user to enter phone number → sends SMS OTP → verifies code → proceeds to CONFIRM .	Wi-Fi traffic, Web traffic, or markets without Header Enrichment gateways.
BOTH (Recommended)	Hybrid: Tries Header Enrichment first. If MSISDN resolved → skips to CONFIRM . If MSISDN missing → gracefully falls back to OTP page.	Universal campaigns handling both 4G/5G Cellular Data and Wi-Fi users seamlessly.
NONE (Direct Redirect)	Direct 1-Click Null-Flow: Immediately appends click_id / attribution parameters and redirects to an external CG URL (cgRedirectUrl).	Pure affiliate routing or external carrier landing gateways.

4. Step 3: Partner API Configuration (**api_configs**)

Configured per campaign in the **API Configuration Tab** ([backend/src/modules/api-config/entities/api-config.entity.js]

(file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/api-config/entities/api-config.entity.js **#L19-L61**)):

4.1 Endpoints Setup

1. **Blocklist Check API (**blocklistApi**)**: Checks if phone number is DND/Blacklisted before billing.
2. **Active Subscription API (**subscriptionApi**)**: Checks if user is already an active subscriber. If active, prevents re-charging and routes directly to **THANKYOU** page

with status **ALREADY_SUBSCRIBED**.

3. **Charge / Subscribe API (**subscribeApi**)**: Triggers actual billing setup on partner gateway.
4. **Header Enrichment Lookup (**resolveMsisdnUrl**)**: External MSISDN lookup endpoint for non-header network gateways.
5. **Headers JSON (**headersJson**)**: Custom authentication headers (e.g. `{"Authorization": "Bearer TOKEN"}`) passed to partner endpoints.

4.2 Dynamic URL Template Placeholders

You can use standard variable placeholders in partner URLs. The engine ([partner-api.service.js:L28-L40](file:///Users/abhinavishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/flow/partner-api.service.js #L28-L40)) automatically interpolates them:

- `{{phone}}` or `{{msisdn}}` → Subscriber mobile number.
- `{{serviceId}}` → Campaign service identifier.
- `{{country}}` / `{{operator}}` → Country and carrier names.
- `{{planId}}` / `{{pack}}` → Selected plan (**daily**, **weekly**, **monthly**).
- `{{subServiceId}}` → Auto-mapped pack code (**HDaily**, **HWeekly**, **HMonthly**).

5. Step 4: Visual Page Builder (GrapesJS Canvas)

5.1 The 6 Standard Campaign Pages

Each campaign consists of standard pages managed in the Canvas Editor ([TemplateEditor.tsx](file:///Users/abhinavishwakarma/work/JPL/wapManager/html-canvas/frontend/src/editor/TemplateEditor.tsx)):

1. **HOME (Landing Page)**: Main promotional offer, hero image, subscription trigger button.
2. **OTP (Verification Page)**: Phone input, OTP code input, "Get OTP" & "Verify & Continue" buttons.

3. **CONFIRM (Billing Confirmation Page)**: Pack picker (Daily/Weekly/Monthly) and final confirm button.
 4. **THANKYOU (Success Page)**: Confirmation message, content access link.
 5. **BLOCKED (Restricted User Page)**: Displayed when partner API returns DND/Blocked status.
 6. **ERROR (Fallback Page)**: Displayed when technical errors or invalid parameters occur.
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6. Step 5: Buttons, Hotspots & Priority Flow Chains

6.1 Visual Button & Hotspot Setup (No Manual Coding Needed!)

When the client selects any button or draws a transparent hotspot overlay in the GrapesJS editor, the right-hand **Property Panel** ([PropertyPanel.jsx:L1088-L1138] (file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/frontend/src/editor/shell/PropertyPanel.jsx #L1088-L1138)) provides visual options:

```
<!-- Automatically generated by Editor UI dropdown selection -->
<button type="button" data-action="SUBSCRIBE" class="flow-btn">Subscribe
Now</button>
<button type="button" data-action="CONFIRM" class="flow-btn">Confirm
Payment</button>
```

Action Dropdown Choices in Editor:

- **"Continue verification flow (HE / OTP)"** → Binds **data-action="SUBSCRIBE"**.
 - **"Another page in this campaign"** → Direct page link (**OTP**, **CONFIRM**, etc.).
 - **"Another website (URL)"** → External URL redirect.
 - **"Another part of this page (Scroll)"** → Smooth scroll to section (**#anchor**).
 - **"Sequential Action Chain (Priority Flow)"** → Advanced multi-step priority chain.
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6.2 Advanced Feature: Priority Action Chains (`data-action="CHAIN"`)

For complex campaigns requiring preliminary API checks before advancing, the client can configure **Sequential Priority Chains** directly in the editor ([PropertyPanel.jsx:L453-L479](file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/frontend/src/editor/shell/PropertyPanel.jsx #L453-L479)):

1. **Priority 1 (Webhook / API Check):** Hits a custom validation URL (e.g. `https://client-partner.com/api/eligibility-check`).
2. **Priority 2 (Page Transition):** If Priority 1 succeeds, proceeds to **CONFIRM** or **OTP** page.

Tip

Why Priority Chains? Allows clients to run custom fraud checks, balance checks, or external CRM logging before allowing the user to enter the billing stage.

7. Step 6: Public Consumer Runtime & Shadow DOM Isolation

7.1 Why Shadow DOM?

([SubscriptionPage.jsx:L89-L94](file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/frontend/src/pages/SubscriptionPage.jsx #L89-L94))

Public subscriber templates are mounted inside a **Shadow DOM Container** (`attachShadow({ mode: 'open' })`):

- **Reason 1 (Style Isolation):** Prevents global application styles from breaking custom client HTML/CSS templates, and vice versa.

- **Reason 2 (Click Interception):** Uses `composedPath()` event traversal to intercept click events on buttons/hotspots without requiring full page reloads.
 - **Reason 3 (Client-Side Navigation):** Router navigation updates query parameters (`step=CONFIRM`) in the browser URL for instantaneous page rendering.
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8. Step 7: Telemetry Analytics, Logs & Session Timelines

8.1 Campaign Analytics Dashboard

(`[analytics.service.js]`

(`file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/analytics/analytics.service.js`))

Tracks metrics per campaign:

- **Total Impressions & Unique Visits.**
- **Conversion Metrics:** `VISIT`, `BLOCKED`, `PLAN_VIEW`, `CONFIRM_VIEW`, `SUBSCRIBE_CLICK`, `SUBSCRIBE_SUCCESS`, `SUBSCRIBE_FAILED`.
- **OTP Analytics Dashboard:** Hourly/Daily OTP delivery rates, resend attempt rates, and provider performance comparison.

8.2 Campaign Event Logs & Elasticsearch

(`[search.service.js]`

(`file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/backend/src/modules/search/search.service.js`))

Searchable real-time logs page (`/logs`) allows filtering by:

- Visit ID, MSISDN, Click ID, Vendor ID, Affiliate ID.

8.3 Session Timeline Modal

([SessionTimelineModal.jsx]

(file:///Users/abhinavvishwakarma/work/JPL/wapManager/html-canvas/frontend/src/components/dashboard/SessionTimelineModal.jsx))

Clicking on any Visit ID opens a visual **User Journey Timeline**:

1. **VISIT** → User landed on HOME page (Timestamp, IP, User-Agent).
2. **HE_RESOLVE** → Header Enrichment MSISDN detected.
3. **BLOCKLIST_CHECK** → Passed DND filter.
4. **SUBSCRIBE_SUCCESS** → Billed successfully on partner gateway.

9. Frequently Asked Questions (Client Q&A Cheat Sheet)

Q1: "Can phone numbers come in any international format?"

Answer: Yes. The system imposes no hardcoded 10-digit or country-specific regex restrictions. Numbers from any country (UAE **971...**, KSA **966...**, Europe, Africa) are sanitized of non-numeric characters and passed directly to partner billing APIs.

Q2: "What if Header Enrichment fails when a user is on Wi-Fi?"

Answer: Under **BOTH** mode, if Header Enrichment fails or returns empty, the engine automatically falls back to the **OTP** page so the user can enter their mobile number manually.

Q3: "Does the client need to write HTML attributes manually for buttons?"

Answer: No. In the GrapesJS visual editor, selecting any button or hotspot opens the Property Panel dropdown where the client selects the desired action visually.

Q4: "How does the system prevent double-charging an already subscribed user?"

Answer: At the **CONFIRM** transition step, `partnerApiService.checkSubscription()` is invoked. If the partner API returns **active**, the engine skips billing and routes the user to **THANKYOU** with status **ALREADY_SUBSCRIBED**.

10. Summary Checklist for Creating a New Campaign

- ☒ **Create Campaign:** Set Name, Country, Operator, Service ID, and Verification Mode (**BOTH**).
- ☒ **Configure Partner APIs:** Fill in `blocklistApi`, `subscriptionApi`, `subscribeApi`, and `headersJson` in API Configuration Tab.
- ☒ **Design Pages in Canvas:** Customize **HOME**, **OTP**, **CONFIRM**, **THANKYOU** templates in GrapesJS Editor.
- ☒ **Verify Button Actions:** Ensure buttons are linked to **SUBSCRIBE** / **CONFIRM** via Property Panel dropdown.
- ☒ **Test Flow Link:** Test entry link `http://localhost:5173/subscription?country=<Country>&operator=<Operator>&campid=<ID>&msisdn=<TestNumber>`.
- ☒ **Audit Telemetry:** Verify visit footprints and logs on `/logs` and `/analytics` dashboards.